

Elisabeth Rebboah, PhD

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Research Profile

Experimental and computational biologist developing multimodal approaches to study rare cells in human disease. Led large-scale experimental and analytical work within ENCODE and IGVF, from study design and data generation through biological interpretation and public release. Current research integrates molecular and clinical data to understand systemic mastocytosis.

Core Expertise

- **Multimodal genomics:** Design and analysis of single-cell RNA-seq, ATAC-seq, spatial transcriptomics, and immune repertoire data in R and Python.
- **Assay development:** Droplet-based, split-pool, combinatorial barcoding, imaging-based, and long-read sequencing workflows.
- **Reproducible computation:** Scalable pipelines using Snakemake, Nextflow, Git, and high-performance computing.
- **Translational research:** Integration of clinically annotated cohorts, biospecimens, molecular assays, and patient outcomes.
- **Scientific leadership:** Cross-functional project leadership, production planning, protocol development, mentorship, and public data release.

Research Experience

Postdoctoral Fellow

September 2025 – present

Advisor: Dr. Hussein A. Abbas

The University of Texas MD Anderson, Houston, TX

- Lead multimodal studies of systemic mastocytosis integrating spatial transcriptomics, single-cell sequencing, proteomics, immune repertoires, and longitudinal clinical data.
- Develop analytical approaches to characterize mast cell states, tissue organization, and variation across disease subtypes and treatment regimens.
- Connect molecular measurements with symptoms and disease severity to identify clinically informative biomarkers and biological mechanisms.
- Build experimental models, collaborative studies, and biobank infrastructure supporting a broader translational program in mast cell disease.

Postdoctoral Scholar

May 2024 – August 2025

Advisor: Dr. Ali Mortazavi

University of California, Irvine, CA

- Led production and analysis teams for more than 25 large-scale single-nucleus RNA-seq experiments spanning brain regions, skeletal and cardiac muscle, liver, kidney, and endocrine glands.
- Characterized cell-type and cell-state variation across genetically diverse mouse strains for the Impact of Genomic Variation on Function consortium.
- Integrated RNA and chromatin accessibility measurements across 10x, Parse Biosciences, and SHARE-seq platforms.
- Built and maintained a scalable Snakemake pipeline for single-cell and single-nucleus RNA-seq processing.
- Mapped spatial organization in the mouse hypothalamus and pituitary by integrating MERSCOPE imaging with whole-transcriptome single-nucleus RNA-seq.

Graduate Student Researcher

Jan. 2019 – April 2024

Advisor: Dr. Ali Mortazavi

University of California, Irvine, CA

- Developed LR-Split-seq, a long-read single-cell RNA-seq assay using split-pool barcoding, and benchmarked it against bulk long-read and short-read single-cell sequencing.
- Identified cell-type and cell-state regulatory programs across five tissues during mouse postnatal development using single-cell RNA-seq and topic modeling.
- Led experimental production, sample organization, analysis, and public data release for large-scale ENCODE single-cell studies.
- Submitted and curated more than 500 transcriptomic and epigenomic experiments for ENCODE, including 235 microRNA-seq datasets and more than 280 single-cell RNA-seq and ATAC-seq datasets.
- Validated single-cell transcriptional states using RNA FISH in differentiating skeletal muscle cells.

Research Associate II

Jan. 2017 – Aug. 2018

Sanford Burnham Prebys Medical Discovery Institute, San Diego, CA

- Conducted CRISPR and RNAi studies in innate immunity and virology.

- Managed high-throughput RNAi screens using automated liquid-handling systems.
- Processed primary human blood samples and supported the development and optimization of cell-based experimental workflows.
- Analyzed next-generation sequencing data generated from functional genomics screens.

Research Associate I

Sep. 2015 – Jan. 2017

BioLegend, San Diego, CA

- Performed quality control testing of recombinant proteins using cell-based and binding assays.
- Maintained primary cell cultures and a large repository of mammalian cell lines.
- Produced audit-ready experimental documentation within a regulated commercial environment.

Education

PhD, Mathematical, Computational, and Systems Biology: University of California, Irvine (2024)

BS, Bioengineering: University of California, San Diego (2015)

Peer-Reviewed Publications

1. **Rebboah, E.**, Weber, R., Abdollahzadeh, E., et al. (2026). Systematic cell-type resolved transcriptomes of 8 tissues in 8 lab and wild-derived mouse strains captures global and local expression variation. *Cell Genomics*, 6(4).
2. Loving, R. K., Sullivan, D. K., Reese, F., **Rebboah, E.**, et al. (2025). Long-read sequencing transcriptome quantification with Ir-kallisto. *PLOS Computational Biology*, 21(12).
3. IGVF Consortium (2024). Deciphering the impact of genomic variation on function. *Nature*, 633(8028).
4. **Rebboah, E.***, Reese, F.*, et al. (2021). Mapping and modeling the genomic basis of differential RNA isoform expression at single-cell resolution with LR-Split-seq. *Genome Biology*, 22(286).
5. Pieraut, S., Gounko, N., Sando, R., Dang, W., **Rebboah, E.**, et al. (2014). Experience-Dependent Remodeling of Basket Cell Networks in the Dentate Gyrus. *Neuron*, 84(1).

*Co-first authors.

Preprints

1. Weber, R., Carilli, M., **Rebboah, E.**, et al. (2026). Hybrid crosses reveal a cell-type-specific landscape of mouse regulatory variation. *bioRxiv*.
2. Swarna, N.P., Boeshaghi, A.S., **Rebboah, E.**, et al. (2025). Determining gene specificity from multivariate single-cell RNA sequencing data. *bioRxiv*.
3. **Rebboah, E.**, Rezaie, N., Williams, B. A., et al. (2024). The ENCODE mouse postnatal developmental time course identifies regulatory programs of cell types and cell states. *bioRxiv*.
4. Rezaie, N., **Rebboah, E.**, Williams, B. A., et al. (2024). Identification of robust cellular programs using reproducible LDA that impact sex-specific disease progression in different genotypes of a mouse model of AD. *bioRxiv*.
5. Reese, F., Williams, B. A., Balderrama-Gutierrez, G., Wyman, D., Celik, M. H., **Rebboah, E.**, et al. (2023). The ENCODE4 long-read RNA-seq collection reveals distinct classes of transcript structure diversity. *bioRxiv*.
6. Balderrama-Gutierrez, G., Liang, H. Y., Rezaie, N., Carvalho, B., Forner, S., Matheos, D., **Rebboah, E.**, et al. (2021). Single-cell and nucleus RNA-seq in a mouse model of AD reveal activation of distinct glial subpopulations in the presence of plaques and tangles. *bioRxiv*.
7. Carvalho, B., **Rebboah, E.**, et al. (2021). Uncovering the Gene Regulatory Networks Underlying Macrophage Polarization Through Comparative Analysis of Bulk and Single-Cell Data. *bioRxiv*.

Selected Presentations

- American Initiative in Mast Cell Diseases Conference, 2026: Plasma Proteomics Identifies a Circulating Signature of Disease Severity in Systemic Mastocytosis
- ASHG Annual Meeting, 2024: Exploring the impact of genetic diversity on cell type- and state-specific gene expression in mice
- ASHG Annual Meeting, 2022: The ENCODE mouse postnatal developmental time course identifies regulatory signatures of cell type maturation

Teaching Experience

COSMOS, University of California, Irvine

- Instructor, Genes, Genomes, and Skeletal Muscle Dystrophies (2021–2023)
- Teaching Assistant, Tissue, Tumor Biology, and Mathematical Modeling (2019)

Teaching Assistant, University of California, Irvine

- Introduction to Precision Medicine (2021–2022)